DE.AUTO

CHAPTER 3

AIR & FUEL SYSTEMS

THE WORLD OF CAR MODIFICATIONS MADE SIMPLE

Cars explained in plain, friendly English

MEMBER EXCLUSIVE • DE.AUTO •

POWERED BY DE.AUTO

DE.AUTO — Because knowledge is power, and cars are passion.

CAR MODIFICATIONS GUIDE MUST-KNOW TERMS

Car modifications, explained

Chapter Three: Air and Fuel Systems

“AIR IN, FUEL IN, POWER OUT.”

— DE.AUTO

Important notice

This guide is **entertainment only**. It is not educational or scientific content, and it is not a how-to manual. DE.AUTO is not an accredited scientific, engineering or technical authority. Everything here is simplified for fun and general interest, and it can differ from one car to another.

You must **never** try any modification, drive at high speed, or do any kind of stunt or dangerous driving on public roads. Safety and respecting traffic laws are everyone's own responsibility.

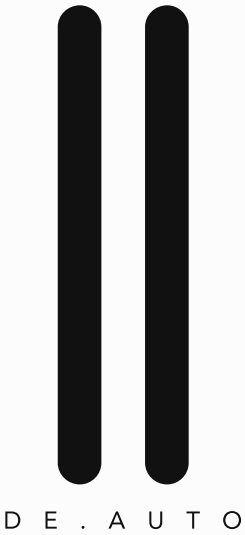
The information here is **general and approximate**, and much of it describes only the typical case. Laws, regulations and specifications differ by country and model, so always check the owner's manual and the official authorities.

Vehicle modifications are subject to local laws. Any unauthorized modification can lead to legal liability, fines or having your vehicle impounded.

DE.AUTO is not responsible for anyone who applies the information in this guide in real life.

Images and drawings in this guide are **illustrative** only. Any company names and trademarks mentioned belong to their owners, and there is no affiliation, sponsorship or endorsement.

For any decision about your car, go to certified professionals and the manufacturer's manual, and only try something if it complies with official regulations and legal requirements. 🚀



DE.AUTO is more than a name. It's an identity that brings the passion for cars together with knowledge and progress.

At DE.AUTO we make fresh content for car lovers, mixing fun and know-how from the world of modifications and high performance, in a simple style that's easy to follow. Our goal is to help you understand your own car better and make smarter calls on modifications and maintenance.

We walk you through the key terms and parts of the modification world, so you can chat with the enthusiasts and follow every detail without feeling like an outsider.

Whether you're a beginner or a pro, you'll find terms, basics and fun facts here that help you get into cars in an organized, confident way.

DE.AUTO — Because knowledge is power, and cars are passion.

READY?

How many litres of air does your engine need to burn a single litre of petrol? 🤯

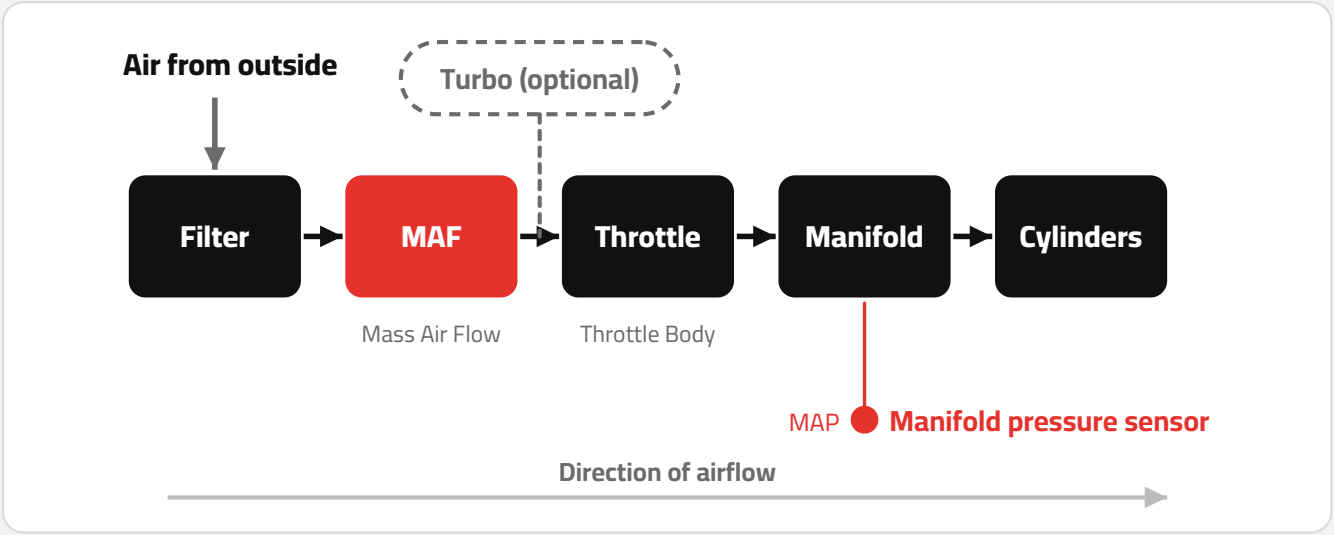
By the end of this chapter you'll know:

- ✓ How does air travel from the atmosphere to the cylinder, and what keeps watch on it along the way?
- ✓ How does petrol get from the tank to the injector, and what is the difference between port and direct injection?
- ✓ What is the ideal air-to-fuel ratio (14.7:1), and when is an engine “rich” and when is it “lean”?
- ✓ What does octane mean, and why is “follow the manufacturer’s recommendation” smarter than “always fill with 95”?
- ✓ How did the carburettor work, and why does it still have fans?
- ✓ What is E85, and when does it make sense and when does it not?
- ✓ What air and fuel mods actually help, and what is just talk?



First: the air's journey 💨

The engine doesn't only "drink" petrol, it drinks far more air. Combustion needs oxygen, and oxygen comes from the air. Between the atmosphere and the cylinder there are five stops:



Simplified drawing. The order of the parts can differ from car to car

Air filter

It stops dust and dirt from entering the engine. In our dusty environment, its job matters a lot.

Throttle body

A gate that opens and closes to control how much air enters the engine.

Intake manifold

A box and pipes that share the air out among the cylinders, and the designer tries to deliver a similar amount to each one.

Sensors (MAF / MAP)

They measure the incoming air and tell the computer, which calculates from them how much fuel to spray.

≈ 9,000 litres of air, roughly, is what the engine needs to burn one litre of petrol completely. The reason: every kilo of petrol needs about 14.7 kilos of air, and air is very light!

🎛️ **The pedal isn't a cable:** In most modern cars the pedal doesn't pull a cable to the throttle. It is a **sensor** that sends a signal to the computer, and the computer decides how far to open the throttle. It is called Drive-by-Wire or the electronic throttle. So pressing the pedal is a "request", not a direct command.

The engine’s eyes: air sensors 👀

The computer **can’t see** the air. It relies on sensors that tell it how much air came in. If a sensor is wrong, all the fuel calculations go wrong.

Airflow sensor (MAF — Mass Air Flow)
A hot wire or thin film inside the airflow. The more air passes, the more it cools, and from the amount of cooling the computer knows the air mass. It is usually located after the filter.

Manifold pressure sensor (MAP — Manifold Absolute Pressure)
Measures the pressure inside the manifold. The computer combines this reading with engine speed and air temperature to estimate the amount of air. In turbo cars it also shows you how much “boost” is going in.

🔎 **Which one is in my car?** Some cars rely mainly on MAF, others on MAP, and some use both. What matters to you: either way, the computer is being told “how much air do I have?”.

The clogged filter: the real story

✕ MYTH
A clogged filter makes the car burn noticeably more petrol.

✓ FACT
In modern cars with fuel injection and a computer, US government tests found no significant difference in fuel economy. The computer adjusts the fuel to the incoming air. What actually suffers is **acceleration**. In older carburettor cars, the effect on economy was more noticeable.

Change the filter on the manufacturer’s schedule, and sooner if you drive in dust or on dirt roads.

A filter isn’t for permanent cleaning: A normal paper filter gets replaced. Don’t beat it clean and put it back until it falls apart.

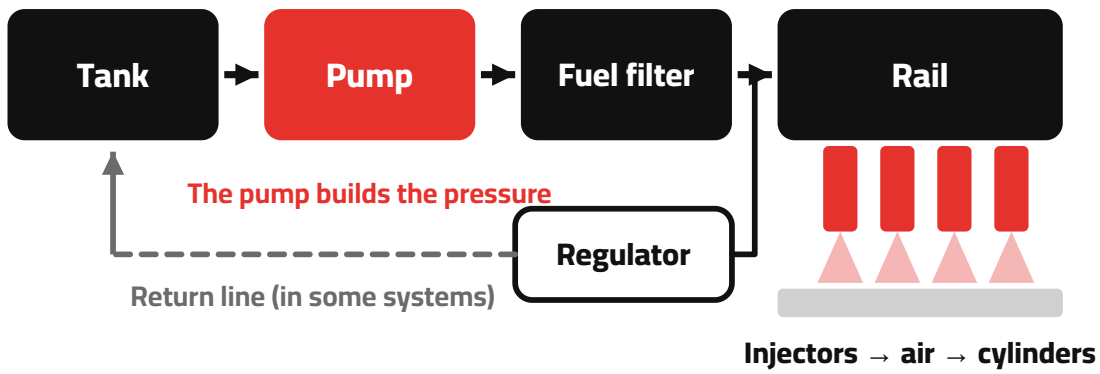
Don’t touch the sensor element with your fingers or a brush. If it needs cleaning, use only a dedicated air-sensor cleaner (MAF Cleaner).

🎯 Challenge 1 **Which sensor measures the air mass entering, using a hot wire or film?**

A) Oxygen sensor | B) MAF sensor | C) Coolant temperature sensor

Second: the fuel's journey ⛽

Fuel doesn't leave the tank by itself. A pump pushes it, a filter cleans it, a regulator sets its pressure, a rail distributes it, and injectors spray it. Focus on the drawing, because every box has a job:



The fuel's journey from the tank to the injectors, simplified

Fuel pump

Electric, and usually inside the tank. It draws the fuel and pushes it at a set pressure to the rest of the system.

Fuel filter

Traps contaminants before they reach the tiny injectors. A clogged one strains the pump and weakens the flow.

Fuel pressure regulator

Holds the pressure steady in the system so the injector delivers the same amount every time.

Fuel rail

A pipe that holds the injectors and delivers fuel to them evenly at constant pressure.

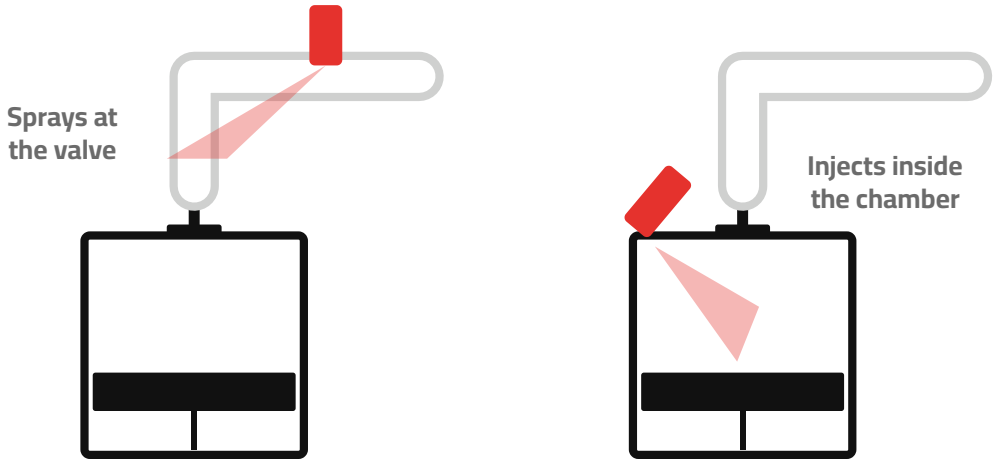
Injectors (Fuel Injectors)

Electric valves that open and close incredibly fast. The computer controls the **opening time** (in fractions of a second), and the longer it is, the more fuel goes in. Injector size is measured by flow rate, such as "cc/min" (cubic centimetres per minute).

🌡️ **Why is an empty tank not good?** The pump inside the tank is **cooled by the fuel** around it. That is why many technicians advise against running with the tank near empty for long stretches as a habit, especially in our heat.

Port or direct injection? 💉

Many people think the injector always “pumps fuel into the cylinder”. That is true for only one type. There are two main types, and which one you have depends on your car:



A simplified cross-section of a single cylinder

Port fuel injection

The injector sprays fuel into the air passage before the valve, so it mixes with the air and they enter together. The pressure is relatively low, about 3–4 bar. The spray also keeps washing the valves as it goes.

Direct injection (GDI)

The injector injects fuel inside the cylinder itself, at much higher pressure (from about 100 to 200 bar or more). It is more precise, and the fuel evaporating inside the chamber cools the mixture, which helps resist knock.

≈ 50X is roughly the pressure difference: from about 3 bar in port injection to about 150 bar in a typical direct injection. That is why direct injection needs a high-pressure pump driven by the engine, in addition to the tank pump.

✕ MYTH

Direct injection is better than port injection in every way.

✓ FACT

It has clear advantages in efficiency and power, but fuel doesn’t pass over the valves, so it doesn’t wash them, and carbon deposits may build up on them over time. That is why some manufacturers combine both types in the same engine (Dual Injection).

When the fuel pump weakens... ⚠️

A weak pump can't deliver the required fuel pressure and volume, especially when you put your foot down and demand rises. The engine feels the hunger before you do!

Weak power and speed
The car feels like it is "choking" when you push hard or climb a hill, but drives normally at gentle speeds.

Hesitation and stuttering
The engine stutters or hesitates under acceleration or at high speeds.

Hard starting
The starter cranks longer than usual before the engine fires.

A loud whine
A louder-than-usual whine from the tank area.

Cutting out or stalling suddenly
In severe cases the car stops or won't start at all.

Check-engine light
The car may log a fault code for fuel pressure or a lean mixture.

❌ **Don't blame the pump before checking other things.** The same symptoms can come from a clogged fuel filter, a fuse or relay, or fuel with water or contaminants in it. The right check is with a fuel pressure gauge.

The logical check sequence

1. Listen for the pump the moment you switch the ignition on, without starting the engine.
2. Ask the workshop to measure the fuel pressure and compare it with the manufacturer's value.
3. Check the fuel filter, the fuse and the relay.
4. Only then can the pump itself be judged.

➡️ **Why does it matter?** When the computer finds the mixture "lean" because of a fuel shortage, it tries to compensate by lengthening the injector opening time, and past a certain limit it can't. And that brings us to the most important point in the whole chapter: **the air-to-fuel ratio** 👇

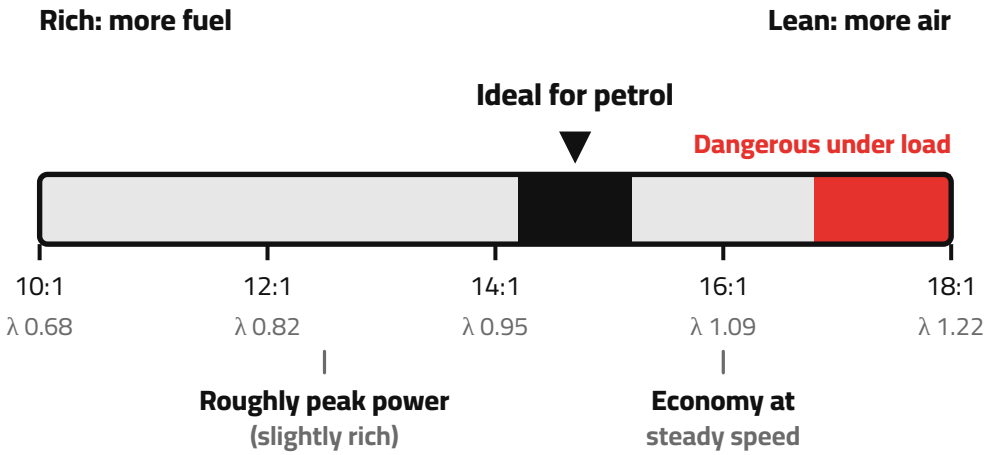
🎯 Challenge 2 **What is the job of the fuel rail?**

A) Cools the pump
B) Stores reserve fuel
C) Distributes fuel to the injectors at constant pressure

Third: the big secret... the air-fuel ratio 🧪

What matters isn't how much air or how much petrol, it is **the ratio between them**. The ratio at which petrol burns almost completely, with no extra air or fuel, is called the ideal ratio (Stoichiometric):

14.7:1 That is, **14.7 kilos of air for every 1 kilo of petrol** (by mass). This number is for regular petrol, and other fuels have a different number.



Figures are approximate and differ from engine to engine. λ = lambda

What is "lambda"? λ

Lambda is an easier way to talk about the ratio without tying it to one particular fuel. You calculate it like this: **actual ratio ÷ ideal ratio**. So:

λ = 1
Ideal: air and fuel perfectly balanced.

λ below 1
Rich mixture: more fuel than needed.

λ above 1
Lean mixture: more air than needed.

Why lambda?
Because the ideal ratio for ethanol, for example, differs from petrol, but λ = 1 means "ideal" in both cases.

⚖️ **Did you know?** The ideal ratio changes with the fuel: regular petrol is about 14.7:1, petrol with 10% ethanol about 14.0, and E85 is around 9.8:1.

Rich or lean? Each has its own story

Sometimes the engine deliberately runs away from ideal, and sometimes it is a fault. The difference is the cause and the duration:

🔴 Rich

More fuel than needed.
Symptoms: petrol smell in the exhaust, dark smoke, dirty spark plugs, higher fuel consumption.
Effect: wasted fuel and carbon build-up, and it can strain the catalytic converter if it goes on for long.

🔵 Lean

More air than needed.
Symptoms: hesitation and stuttering, unstable idle speed, sometimes popping back through the intake.
Effect: higher combustion temperature, and that is where the real danger lies 👇

🔥 **Why is lean under load dangerous?** When you push hard and ask for maximum power, the temperature and pressure inside the cylinder rise. The fuel is what **cools** the chamber (it evaporates and pulls heat away). If the mixture is lean at exactly that moment: higher heat + less cooling = a tendency to knock. Carrying on like that can damage the piston, valves and spark plug. That is why "lean under load" is one of the most dangerous faults.

So why does the engine deliberately run "a little rich" at high performance?

Cooling: The extra fuel evaporates and pulls heat from the chamber and valves.

Safety: It lowers the risk of knock, and protects the valves and the turbo from high exhaust temperatures.

Power: Peak power usually shows up at a mixture slightly richer than ideal (roughly 12.5–13:1 in naturally aspirated engines, and turbo engines are usually richer than that).

Who controls all this? The oxygen sensor

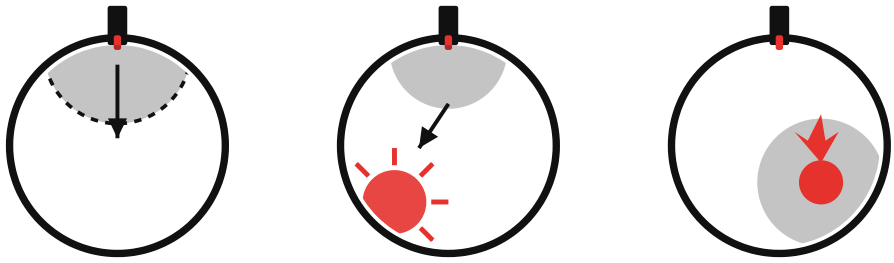
In the exhaust there is a sensor that **reads the leftover oxygen** and tells the computer "is the mixture rich or lean?". The computer corrects it immediately. This is called **closed loop**, and it is used at idle and in normal driving. At full pedal it mostly relies on preset tables (Open Loop). There is also an advanced type called the **wideband** sensor, which gives the actual ratio number, like 12.8:1, and tuners use it on the dyno instead of guessing.

🎯 Challenge 3 **What is the most dangerous mixture condition when you go for maximum power?**

A) A lean mixture | B) Slightly rich | C) Exactly ideal

Fourth: octane and knock 🔊

Not all petrol has the same ability to "withstand". When the mixture is squeezed and the temperature rises, some of it ignites **by itself** before its time. This is the most dangerous thing for an engine:



1. Healthy combustion
The flame spreads calmly from the spark plug outward

2. Knock
The end of the mixture explodes by itself, before the flame reaches it

3. Pre-ignition
A hot spot ignites the mixture before the spark plug fires

A combustion chamber from above (simplified illustration)

Knock (Detonation)

The spark starts combustion normally. But the end of the mixture (the part far from the spark plug) gets squeezed and heated until it ignites by itself before the flame arrives, and the pressure waves collide and you hear a "knocking" sound.

Pre-ignition

The mixture ignites before the spark plug fires, because of a hot spot, such as carbon deposits or an over-hot spark plug. It is usually more dangerous than ordinary knock.

🛡️ **How does the car protect itself?** The computer has a **knock sensor** that listens to the engine. If it hears knock, it retards the spark (reduces the advance) to protect the engine, and as a result you feel your car is weaker. So the car protects itself at the expense of power.

Where does octane come in?

The octane number measures the fuel's resistance to self-ignition (knock). The higher the number, the more pressure and heat the mixture can take before it ignites by itself.

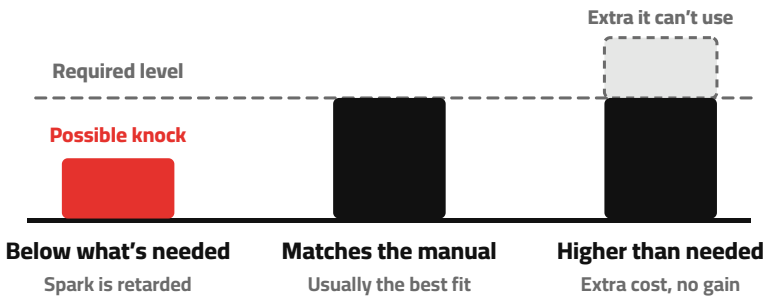
❌ **Common mistake:** Higher octane doesn't mean more energy per litre, or "cleaner", or "stronger" petrol. It only means **higher resistance to knock**.

✅ **DE.AUTO tip:** If you hear constant knock, don't ignore it. Try fuel from another station and go back to the manufacturer's manual, and if it continues, consult a qualified technician. Repeated knock wears the engine out.

91 or 95? The golden rule 🏅

The common line is: 95 for powerful cars and turbos, and 91 for regular ones. That shortcut isn't enough. The correct rule:

THE MANUAL is the judge. **Follow the manufacturer's recommendation** for your engine. They tested your engine on those numbers, and it is usually written in the manual and on the fuel filler flap.



Column height = the fuel's resistance to knock. The drawing is illustrative

Why do some engines want higher octane?

High compression ratio: the mixture is squeezed more, so it gets hotter.

Turbo and supercharger: compressed air raises the pressure and heat inside the chamber (Chapter 7).

📖 **Watch the wording:** Read the wording in the manual: **"required"** is different from **"recommended"**. The first means don't go below this number. With the second, the engine usually adapts to lower, but performance may drop. And if the manual allows more than one number, the lower is the minimum that is safe.

✕ MYTH
Higher-octane petrol makes any car more powerful.

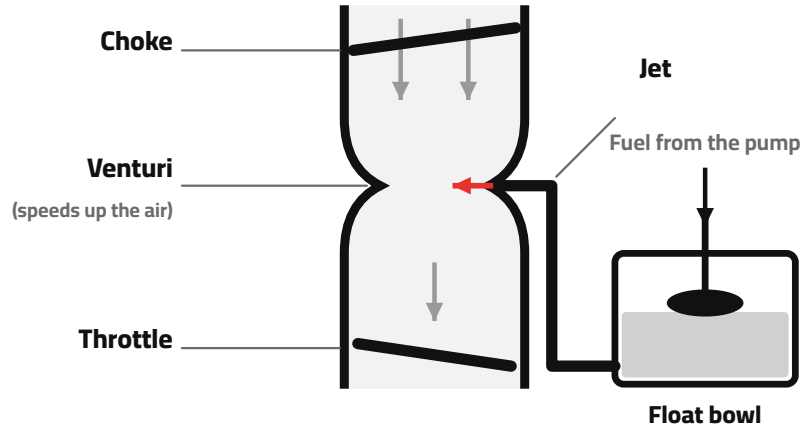
✓ FACT
Power only increases if the engine is designed or tuned to benefit from it. A regular car that doesn't knock gets no benefit. And at our pumps there are numbers like 91, 95 and 98. Choose by the manual, not by "the priciest".

🎯 **Challenge 4** **Your car's manual asks for 91 petrol. You filled up with 98. What is the likely result?**

A) Usually no difference, and you pay more
B) Noticeably more power and speed
C) The engine breaks

Fifth: the carburettor... grandfather of fuel injection 🕰️

Before injectors and computers existed, fuel was mixed with air by simple, beautiful physics, with not a single sensor. This is the carburettor:



A simplified carburettor (a top-to-bottom model)

1. Air flows through the tube, and in the middle the passage narrows (the venturi), so the air speeds up and its pressure drops.
2. The lower pressure draws fuel out of the jet (a small calibrated opening) from a bowl with a float that holds the fuel level steady.
3. Air and fuel mix and pass through the throttle to the engine, and the pedal opens the throttle.
4. The choke blocks some of the air during a cold start, to enrich the mixture.

❌ Why did injection replace it?

The carburettor has no computer or oxygen sensor, so it can't keep the ratio close to ideal. It is affected by temperature, altitude and the car's tilt, needs a choke for cold starts, and struggles to meet modern emissions rules. Injection is more precise, more economical and easier in all conditions.

❤️ Why does it still have fans?

Purely mechanical: you can see everything with your own eyes and tune it by hand with the jets. Its sound and feel are part of the charm of classic cars. But it needs experience and patience, and doesn't give you the comfort of injection.

🌱 **Did you know?** The carburettor still exists in some small equipment, like some small motorbikes and garden machines, but new cars use fuel injection.

✅ **For classic lovers:** If you buy a classic car with a carburettor: don't compare it with your modern car. It needs periodic tuning, specialist parts, and a technician who knows how to work on it.

Sixth: E85 and flex fuel 🌽

In the modification world you hear: "E85 makes more power". Let's see the whole truth, the sweet and the bitter:

E85 is a fuel containing ethanol at up to 85% blended with petrol (the actual percentage varies). Cars that can run on it are called **flex fuel** vehicles, and they carry a sensor that measures the ethanol percentage in the fuel, and the computer adjusts accordingly.

Property	Regular petrol	E85
Knock resistance	Normal (depends on the number)	Clearly higher than 98 (roughly 100 to 105)
Ideal ratio	14.7:1	About 9.8:1
Energy per litre	Baseline	About a quarter less
Fuel needed for the same power	Baseline	Roughly 30–40% more
Fuel economy	Baseline	Worse (uses more for the same distance)

The sweet side

Its high knock resistance lets the tuner raise the spark advance and boost in a prepared engine, giving higher performance. And its evaporation cools the mixture.

The bitter side

Higher consumption, and it needs more fuel flow (capable injectors and pump), plus essential tuning. And in cold weather starting can be hard.

❌ **Don't fill with E85 in a car that isn't prepared for it.** The rubber, seals and fuel-system parts in ordinary cars aren't designed for a high percentage of ethanol, and the computer doesn't know how to handle it. Not every car can.

🛰️ **Honest from us: Is E85 available here?** What is officially announced by the largest fuel supplier in the Kingdom: 91, 95 and 98 (unleaded petrol). **We found no reliable source confirming that E85 is generally available at stations in the Kingdom**, so don't assume it is. Ask the company and the station before any planning.

🔜 Tuning for Flex Fuel and how the computer changes the mixture according to the ethanol percentage is explained in **Chapter 12: Car Tuning and ECU**.

🎯 Challenge 5 **Why does E85 need bigger injectors and more fuel flow?**

A) Because it is cheaper

B) Because its energy per litre is lower, so we need a larger quantity

C) Because it just evaporates more slowly

Seventh: improving air and fuel 🛠️

Every air and fuel modification plays in one equation, and each link depends on the one before it. These are the best-known mods, honestly:



Sport / high-flow air filter

In theory it lets more air through. On ordinary cars the difference is usually small, and some types reduce filtering efficiency if you choose a poor one or don't maintain it properly.

Cold air intake

Replaces the intake pipe and tries to draw in cooler air. Cold air is denser, so there is more oxygen. But if it is mounted low and you drive through water, water can enter the engine and that is major damage.

High-flow injectors

You only need them when fuel demand rises (a bigger turbo, major mods, E85). And tuning is needed to calibrate them, otherwise the mixture goes too rich.

Catch can

A small tank that catches the oil vapour and moisture coming out of the crankcase ventilation before it returns to the intake. It needs periodic emptying.

System maintenance

Air and fuel filters on the manufacturer's schedule, and cleaning the injectors and throttle body if there are deposits. The cheapest "modification" to start with.

✕ MYTH
A sport filter gives big power.

✓ FACT
Usually a small difference, and sometimes you won't feel it. It has maintenance needs, and deserves a known-quality brand.

✕ MYTH
A cleaning additive gives big power and savings even if the fuel system is already clean.

✓ FACT
It helps when there are real deposits, and restores normal performance. It doesn't add on top of stock, and it doesn't work miracles on fuel economy.

⚖️ **Law first:** Don't vent the crankcase ventilation straight to the air, as that pollutes and is illegal in many countries. And any modification that touches emissions may break regulations and affect the warranty. Check with the dealer and the relevant authorities before installing.

Common mistakes to avoid ❌

❌ **Putting in 98 for a car whose manufacturer says 91, and expecting it to fly.** Usually nothing changes except the bill.

❌ **Ignoring knock** because it "doesn't matter", or pouring random products into the tank instead of diagnosing the problem.

❌ **Touching the air sensor element or cleaning it with an unsuitable product,** which ruins the sensor and its readings.

❌ **Fitting a sport filter and over-oiling it,** so oil leaks onto the sensor and confuses it.

❌ **Fitting bigger injectors and leaving them untuned,** so the mixture runs rich and the engine gets confused.

❌ **Putting E85 in a non-flex-fuel car** because someone told you "it makes more power".

Garage talk vs proper names 🗣️

Gulf garage talk	Proper name	What it means
Throttle	Throttle Body	The air gate that controls the amount
Antik / manifold	Intake Manifold	The box that distributes the air
Bakhakhat / rashashat	Fuel Injectors	What sprays the fuel
Tarumba	Fuel Pump	What pushes the fuel from the tank
Reel	Fuel Rail	A pipe that distributes the fuel
Khalta ("the mix")	Air-Fuel Mixture	What actually burns in the cylinder
Rich / lean engine	Rich / Lean	More fuel / more air than needed
Taqtaqa / daqq	Knock / Detonation	Early self-ignition of the mixture
Oxygen sensor	O2 / Lambda Sensor	Measures the oxygen in the exhaust
Air sensor	MAF Sensor	Measures the mass of incoming air
Karbureter	Carburettor	An old mechanical mixing system
Katch kan	Catch Can	A tank that catches oil vapour

Wrap-up 📌

Now you know the whole journey: air from the filter to the cylinder, fuel from the tank to the injector, the ideal ratio 14.7:1, how to choose your petrol correctly, the idea of the carburettor, and E85. And you learned how to tell a useful modification from marketing talk! 🚀

🎯 Challenge answers:

1) B: the MAF sensor.

---

2) C: it distributes fuel to the injectors at constant pressure.

---

3) A: a lean mixture under load (higher heat and less cooling).

---

4) A: usually no difference, and you pay more.

---

5) B: because E85 has less energy per litre, so we need a larger quantity of it.

➡️ But one question is left hanging: the air came in and the fuel burned… where does all of it go? In **Chapter 4: The Exhaust System** we explain the headers, the downpipe and the cat-back, and answer: does more exhaust flow really add power? Don't miss it! 🔥